

IVANHOE, Australia

WMO: 956970

Lat: 32.8831S

Lon: 144.3089E

Elev: 101

StdP: 100.12

Time Zone: 10.00 (AUS)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	2.2	3.4	-8.7	1.8	25.2	-6.8	2.1	23.7	9.5	15.9	8.8	15.8	2.9	210	0.500

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	13.7	41.2	19.5	39.2	19.1	37.3	18.6	22.8	30.5	21.8	30.4	21.0	30.9	5.1	340

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.3	16.1	24.2	19.6	14.5	24.0	18.2	13.2	24.2	67.8	30.4	64.4	29.7	61.3	30.9	26.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
10.5	9.0	7.9	DB	-0.1	43.5	1.1	1.9	-0.9	44.9	-1.5	46.0	-2.1	47.1	-2.9	48.5	(4)
			WB	-0.7	24.5	1.1	1.1	-1.5	25.3	-2.1	25.9	-2.7	26.5	-3.5	27.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	19.1	28.1	27.0	23.3	19.2	14.4	11.3	10.9	12.1	15.7	19.5	22.9	25.5
	DBStd	7.05	4.42	4.36	3.91	3.74	2.99	2.49	2.50	2.73	3.59	4.42	4.68	4.45
	HDD10.0	40	0	0	0	0	1	14	18	7	0	0	0	0
	HDD18.3	945	0	0	5	33	128	211	230	193	97	38	9	2
	CDD10.0	3369	561	475	414	277	136	53	47	73	170	296	388	479
	CDD18.3	1234	303	242	161	59	4	0	0	2	17	76	147	223
	CDH23.3	16580	4367	3249	1848	622	42	1	1	19	276	1042	2004	3109
	CDH26.7	9057	2702	1883	877	209	2	0	0	3	82	471	1060	1769
Wind	WSAvg	4.2	4.7	4.6	4.3	3.7	3.6	3.6	3.9	4.2	4.6	4.6	4.6	4.6
Precipitation	PrecAvg	327	46	19	34	27	35	21	23	28	24	29	22	25
	PrecMax	880	210	121	187	178	96	64	68	68	81	99	120	105
	PrecMin	69	1	0	0	0	0	0	0	3	2	0	0	0
	PrecStd	149	53	27	40	38	28	17	17	20	20	29	23	26
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	44.0	43.4	38.0	34.3	26.5	22.0	22.6	26.0	31.9	36.5	40.9	41.8
		MCWB	20.1	21.4	18.6	17.7	15.1	14.5	13.2	13.2	14.4	16.3	18.4	18.6
	2%	DB	41.6	40.5	35.5	31.0	24.4	19.4	20.1	22.7	29.0	34.1	37.7	39.2
		MCWB	19.7	19.8	18.1	16.8	14.7	12.8	12.1	12.1	14.0	16.0	17.7	18.6
	5%	DB	39.6	37.8	33.5	29.1	22.5	17.8	18.0	20.6	26.6	31.6	35.0	37.2
		MCWB	19.5	19.0	17.9	16.1	13.9	12.2	11.1	11.2	13.2	15.4	17.3	17.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	10%	DB	37.4	35.5	31.6	27.0	20.5	16.3	16.4	18.6	24.0	29.2	32.7	35.1
		MCWB	19.1	19.0	17.7	15.4	13.2	11.6	10.7	10.7	12.4	14.9	16.2	17.6
	0.4%	WB	24.0	24.4	22.8	19.9	18.0	15.6	16.5	15.0	16.6	19.4	21.5	23.3
		MCDB	30.4	31.0	28.0	26.1	20.0	18.3	18.3	19.4	24.1	28.0	31.6	28.3
	2%	WB	22.4	23.0	21.4	18.6	16.1	14.2	13.6	13.5	15.4	18.0	20.3	21.5
		MCDB	31.4	31.8	27.5	24.6	21.4	17.1	17.1	20.1	24.3	27.8	28.3	29.6
Mean Daily Temperature Range	5% DB	WB	21.6	21.9	20.1	17.5	15.0	13.4	12.1	12.2	14.2	17.0	19.3	20.4
		MCDB	32.2	29.7	28.0	24.8	20.2	16.3	16.3	18.8	23.5	27.5	28.2	30.7
	5% WB	WB	20.9	20.9	19.0	16.6	14.0	12.5	11.2	11.3	13.2	15.9	18.1	19.2
		MCDB	32.5	30.9	28.1	25.0	19.2	15.4	15.5	17.3	22.5	27.1	30.0	31.2
Clear-Sky Solar Irradiance	taub	0.332	0.335	0.317	0.304	0.292	0.287	0.286	0.294	0.329	0.326	0.340	0.331	
	taud	2.530	2.549	2.581	2.586	2.590	2.597	2.586	2.552	2.438	2.471	2.471	2.504	
	Ebn at Noon	1006	983	967	927	888	867	886	925	938	980	991	1010	
	Edh at Noon	111	105	95	84	74	69	73	85	108	112	117	114	
All-Sky Solar Radiation	RadAvg	7.93	7.07	6.02	4.60	3.36	2.71	2.99	3.98	5.40	6.66	7.33	8.02	
	RadStd	0.45	0.37	0.36	0.22	0.20	0.15	0.16	0.26	0.30	0.38	0.52	0.30	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (2 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page